Alan Huang

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Education

Stanford University — Stanford, CA

Bachelor of Science in Computer Science — September 2024 – May 2028

- GPA: 3.91 / 4.00

Master of Science in Computer Science (Coterminal) — September 2026 – May 2028

Experience

Software Developer Co-op — January 2027 – May 2027

IBM — *Durham, NC*

- Incoming Software Developer Co-op for Spring 2027, working in a hybrid engineering role at IBM's Durham office

Software Engineer Intern — June 2026 – August 2026

Walmart Global Tech — *Bentonville, AR*

- Engineered and deployed a Python automation service for the Walmart Fulfillment Services (WFS) team that consolidated queries from six backend fulfillment systems, enabling 40 seller-support agents to investigate order status and shipping exceptions 67% faster (15→5 minutes)
- Delivered 8 production bug fixes and authored 15 regression test cases for a Java order-tracking service processing over 500K daily orders, improving system reliability through peer-reviewed pull requests

AI Research Intern — June 2025 – August 2025

National Science Foundation (NSF) — *Indianapolis, IN*

- Improved cross-dataset deepfake detection accuracy from 59.7% to 75.5% on the CelebDF benchmark by designing a diffusion-based Synthetic Domain Alignment technique, validated across FaceForensics++ and CelebDF
- Published as co-first author: *Robust Deepfake Detection via Synthetic Domain Alignment*, 22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (IEEE MASS 2025)

Projects

OS: Crash-Consistent Filesystem & Encrypted Virtual Memory | *C, C++* — 2026

- Implemented a Unix V6 filesystem reader in C with write-ahead journaling and crash recovery, restoring consistency after simulated mid-write failures
- Built a memory-mapped encrypted file system in C++ using the Clock page-replacement algorithm to manage 128 physical memory pages, achieving a 27% reduction in page faults compared to a naive FIFO baseline

Multi-Label Hate Crime Bias Classification | *Python, scikit-learn, XGBoost* — 2026

- Trained and benchmarked five classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost, majority baseline) to predict bias motivation category across 32,508 California hate crime incidents
- Addressed severe class imbalance (three rarest classes <3% of data) using SMOTE and stratified splitting, more than doubling macro F1 from 12.4% (majority-class baseline) to 25.0% with L2-regularized Logistic Regression

Rolyn – Full-Stack Classifieds Platform | *JavaScript, Express, SQLite, Supabase* — 2026

- Built listing detail pages and an authentication backend; implemented secure login using Supabase Auth and JWTs, supporting 50+ user accounts with role-based access across listing and messaging flows

Publications

Alan Huang, Ian Yee, Shu Hu. *Robust Deepfake Detection via Synthetic Domain Alignment.* The 22nd IEEE International Conference on Mobile Ad-Hoc and Smart Systems (IEEE MASS 2025). DOI

Alan Huang, Shaheer Raza, Yuni Xia. *Predicting Diabetes with Medical and Demographic Data.* The 6th International Conference on Computer Science, Engineering, and Education (CSEE 2025). DOI

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL
ML/Data: TensorFlow, scikit-learn, XGBoost, pandas, NumPy, OpenCV
Systems & Tools: Git, Node.js, FastAPI, Express, Supabase, Google Cloud Platform